

3.2 Predicates and Quantified Statements II (Homework)

 INSTRUCTOR

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Due Date: THU, SEP 24, 2026 11:59 PM EDT

 REQUEST EXTENSION

Current Score: – / 100 POINTS | 0.0 %

Scoring and Assignment Information

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QUESTION	1	2	3	4	5	6	7	8	9
POINTS	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.12

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

1. [– / 11.11 Points]

DETAILS

MY NOTES

EppDiscMath5 3.2.004.

Find an informal negation for each of the statements in (a)–(d). Be careful to avoid negations that are ambiguous.

- (a) All dogs are friendly.
- ☐ All dogs are unfriendly.

☐ Not all dogs are unfriendly.

☐ Some dogs are friendly.

☐ There is at least one unfriendly dog.

☐ There is at least one friendly dog.

(b) All graphs are connected.

- ☐ All graphs are disconnected.
- ☐ Not all graphs are disconnected.
- ☐ Some graphs are connected.
- ☐ There is at least one graph that is disconnected.
- ☐ There is at least one graph that is connected.

(c) Some suspicions were substantiated.

- ☐ All suspicions were substantiated.
- ☐ All suspicions were unsubstantiated.
- ☐ Some suspicions were unsubstantiated.
- ☐ There is at least one suspicion that was unsubstantiated.
- ☐ There is at least one suspicion that was substantiated.

(d) Some estimates are accurate.

- ☐ All estimates are accurate.
- ☐ All estimates are inaccurate.
- ☐ Some estimates are inaccurate.
- ☐ There is at least one estimate that is inaccurate.
- ☐ There is at least one estimate that is accurate.

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2. [- / 11.11 Points]

DETAILS

MY NOTES

EppDiscMath5 3.2.008.

Let Statement S be "There are no simple solutions to life's problems."

Which of the following is an informal negation of Statement S?

- ☐ All of life's problems do not have a simple solution.

- ☐ Some of life's problems do not have a simple solution.
- ☐ None of life's problems have a simple solution.
- ☐ At least one of life's problems has a simple solution.
- ☐ At least one of life's problems does not have a simple solution.

Which of the following is a formal version of Statement S ?

- ☐ $\forall$ life problem, x does not have a simple solution.
- ☐ $\exists$ a simple solution x , for each of life's problems.
- ☐ $\forall$ life problem, x has a simple solution.
- ☐ $\forall$ life problem, $\exists$ a simple solution for x .
- ☐ $\exists$ a simple solution x , $\forall$ life problem.

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3. [- / 11.11 Points]

DETAILS

MY NOTES

PRACTICE ANOTHER

EppDiscMath5 3.2.012.

A statement and a proposed negation are given below.

Statement: The product of any irrational number and any rational number is irrational.

Proposed negation: The product of any irrational number and any rational number is rational.

Which of the following correctly describes the relation between the statement and the proposed negation?

- ☐ The proposed negation is correct.
- ☐ The proposed negation is not correct. A possible correct negation would be: There is an irrational number and a rational number whose product is rational.
- ☐ The proposed negation is not correct. A possible correct negation would be: There is not an irrational number and a rational number whose product is rational.
- ☐ The proposed negation is not correct. A possible correct negation would be: There exists an irrational product of an irrational number and a rational number.
- ☐ The proposed negation is not correct. A possible correct negation would be: There does not exist an irrational product of any irrational number and any rational number.

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Let $D = \{-48, -14, -8, 0, 1, 3, 16, 23, 26, 32, 36\}$. Determine which of the following statements are true and which are false. For each statement that is true, enter TRUE. For each statement that is false, enter a value of x that could be used to show the statement is false.

(a) $\forall x \in D$, if x is odd then $x > 0$.

(b) $\forall x \in D$, if x is less than 0 then x is even.

(c) $\forall x \in D$, if x is even then $x \leq 0$.

(d) $\forall x \in D$, if the ones digit of x is 2, then the tens digit is 3 or 4.

(e) $\forall x \in D$, if the ones digit of x is 6, then the tens digit is 1 or 2.

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Consider the following statement.

$\forall$ integer d , if $\frac{6}{d}$ is an integer then $d = 3$.

Which of the following is a negation for the statement?

- ☐ $\exists$ an integer d such that $\frac{6}{d}$ is an integer and $d \neq 3$.
- ☐ $\exists$ an integer d such that $\frac{6}{d}$ is not an integer and $d = 3$.
- ☐ $\forall$ integer d , if $\frac{6}{d}$ is not an integer, then $d \neq 3$.
- ☐ $\forall$ integer d , if $d = 3$, then $\frac{6}{d}$ is an integer.
- ☐ $\forall$ integer d , if $d \neq 3$, then $\frac{6}{d}$ is not an integer.

$\forall$ integer d , if $d \neq 0$, then $\frac{8}{d}$ is not an integer.

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6. [- / 11.11 Points]

DETAILS

MY NOTES

PRACTICE ANOTHER

EppDiscMath5 3.2.027.

Let Statement S be the following.

$\forall$ integer d , if $\frac{8}{d}$ is an integer, then $d = 4$.

Which of the following is the contrapositive of Statement S ?

☐ $\exists$ an integer d such that $\frac{8}{d}$ is not an integer and $d = 4$.

☐ $\exists$ an integer d such that $\frac{8}{d}$ is an integer and $d \neq 4$.

☐ $\forall$ integer d , if $\frac{8}{d}$ is not an integer, then $d \neq 4$.

☐ $\forall$ integer d , if $d \neq 4$, then $\frac{8}{d}$ is not an integer.

☐ $\forall$ integer d , if $d = 4$, then $\frac{8}{d}$ is an integer.

Is the contrapositive true or false? Enter a value for d that can help answer this question.

$d =$

When you substitute this value for d , which of the choices below answers the question?

☐ The contrapositive is true because $\frac{8}{d}$ is not an integer.

☐ The contrapositive is false. *Counterexample* : For the given value of d , $d = 4$ and $\frac{8}{d}$ is not an integer.

☐ The contrapositive is false. *Counterexample* : For the given value of d , $d \neq 4$ and $\frac{8}{d}$ is an integer.

☐ The contrapositive is false. *Counterexample* : For the given value of d , $d \neq 4$ and $\frac{8}{d}$ is not an integer.

Which of the following is the converse of Statement S ?

☐ $\exists$ an integer d such that $\frac{8}{d}$ is not an integer and $d = 4$.

☐ $\exists$ an integer d such that $\frac{8}{d}$ is an integer and $d \neq 4$.

☐ $\forall$ integer d , if $\frac{8}{d}$ is not an integer, then $d \neq 4$.

☐ $\forall$ integer d , if $\frac{8}{d}$ is not an integer, then $d \neq 4$.

☐ $\forall$ integer d , if $d \neq 4$, then $\frac{8}{d}$ is not an integer.

☐ $\forall$ integer d , if $d = 4$, then $\frac{8}{d}$ is an integer.

Is the converse true or false?

☐ The converse is true.

☐ The converse is false.

Which of the following is the inverse of Statement S?

☐ $\exists$ an integer d such that $\frac{8}{d}$ is not an integer and $d = 4$.

☐ $\exists$ an integer d such that $\frac{8}{d}$ is an integer and $d \neq 4$.

☐ $\forall$ integer d , if $\frac{8}{d}$ is not an integer, then $d \neq 4$.

☐ $\forall$ integer d , if $d \neq 4$, then $\frac{8}{d}$ is not an integer.

☐ $\forall$ integer d , if $d = 4$, then $\frac{8}{d}$ is an integer.

Is the inverse true or false?

☐ The inverse is true.

☐ The inverse is false.

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7. [- / 11.11 Points]

DETAILS

MY NOTES

EppDiscMath5 3.2.038.

True or false? All occurrences of the letter u in "Discrete Mathematics" are lowercase. Choose the answer that explains your reasoning.

☐ True. There are no occurrences of the letter u in "Discrete Mathematics", so the given statement is vacuously true.

☐ True. There is one occurrence of the letter u in "Discrete Mathematics" that is lowercase.

☐ False. There is one occurrence of the letter u in "Discrete Mathematics" that is uppercase.

☐ False. There is one occurrence of the letter u in "Discrete Mathematics" that is not lowercase.

☐ False. There are no occurrences of the letter u in "Discrete Mathematics", so the given statement is false.

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8. [- / 11.11 Points]

DETAILS

MY NOTES

EppDiscMath5 3.2.042.

Consider the following statement.

Passing a comprehensive exam is a necessary condition for obtaining a master's degree.

Which of the following correctly expresses this statement in if-then form? (Select all that apply.)

- ☐ If a person cannot obtain a master's degree, then that person passed a comprehensive exam.
- ☐ If a person obtained a master's degree, then that person passed a comprehensive exam.
- ☐ If a person does not pass a comprehensive exam, then that person can obtain a master's degree.
- ☐ If a person passed a comprehensive exam, then that person obtained a master's degree.
- ☐ If a person does not pass a comprehensive exam, then that person cannot obtain a master's degree.

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9. [- / 11.12 Points]

DETAILS

MY NOTES

EppDiscMath5 3.2.046.

Consider the following statement.

Having a large income is not a necessary condition for a person to be happy.

Which of the following expresses the statement without using the words *necessary* or *sufficient*? (Recall that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an *and* statement.)

- ☐ All persons who do not have a large income are not happy.
- ☐ There is a person who does not have a large income and is happy.
- ☐ All persons who have a large income are happy.
- ☐ There is a person who has a large income and is not happy.
- ☐ There is a person who has a large income and is happy.

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